5. Maintenance: Regularly updating and improving the software to fix bugs, add new features, and address security vulnerabilities.

6. Testing: Verifying that the software meets its requirements and is free of bugs.

7. Design Patterns: Solving recurring problems in software design by providing templates for solving them.

8. Agile methodologies: Using iterative and incremental development processes that focus on customer satisfaction, rapid delivery, and flexibility.

9. Continuous Integration & Deployment: Continuously integrating the code changes and deploying them into the production environment.

Main Attributes of Software Engineering

Software Engineering is a systematic, disciplined, quantifiable study and approach to the design, development, operation, and maintenance of a software system. There are four main Attributes of Software Engineering.

1. Efficiency: It provides a measure of the resource requirement of a software product efficiently.

2. Reliability: It assures that the product will deliver the same results when used in similar working environment.

3. Reusability: This attribute makes sure that the module can be used in multiple applications.

4. Maintainability: It is the ability of the software to be modified, repaired, or enhanced easily with changing requirements.

Dual Role of Software

There is a dual role of software in the industry. The first one is as a product and the other one is as a vehicle for delivering the product. We will discuss both of them.